

Generation Interconnection System Impact Study Report

For Queue Project Q12-0158

Mira Loma 500 kV
September 28, 2013

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0158. The Interconnection Customer proposes a 969.9 MW natural gas combined cycle facility interconnecting at the Mira Loma 500 kV substation in the Illinois area. The request was received on September 16, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 121 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$118,670,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of EASY RD to MIRA LOMA 500.0 kV, the MIRA LOMA to YKK USA CHESTNEY 230.0 kV L loads to 148.1% of its 613 MVA rating. Project DFAX: 89.7%.

For the loss of HILLTOP 500/345 kV TX, the TALENENERGY MARTINS CREEK LLC WILLIAMSPORT 500 kV/345 kV Transformer loads to 109.6% of its 821 MVA rating. Project DFAX: 6.2%.

For the loss of ORCHARD ROAD to FT HUMBUG 500.0 kV, the FT HUMBUG to W CAMPUS 345.0 kV line loads to 109.1% of its 1,371 MVA rating. Project DFAX: 8.0%.

For the loss of HILLTOP to TALENENERGY MARTINS CRK LLC WILLIAMSPORT SUBSTATION 500.0 kV line, the HILLTOP 500-345 kV XFMR loads to 108.0% of its 806 MVA rating. Project DFAX: 7.2%.

LITTLE CHUTE to TALENENERGY MARTINS CRK LLC HARRISBURG 345.0 kV L: 105.7% loading (1,012 MVA) following the outage of the CORONET to FORT HUMBUG 500.0 kV; DFAX = 6.7%.

For the loss of EASY RD to MIRA LOMA 500.0 kV, the MIRA LOMA 500/230 kV TX loads to 100.7% of its 1,018 MVA rating. Project DFAX: 99.9%.

BEAVER SUBSTATION to ONANCOCK DIESEL GENERATION FACILITY 230.0 kV L: 100.2% loading (605 MVA) following the outage of the BEAVER SUBSTATION to EUFAULA 500.0 kV Line; DFAX = 11.1%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
MIRA LOMA to YKK USA CHESTNEY 230.0 kV L	Rebuild line with higher-capacity conductor	\$65,778,000	99% (\$65,072,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
TALENENERGY MARTINS CREEK LLC WILLIAMSPORT 500-345 kV Transformer	Replace transformer with higher-rated unit	\$32,577,000	21% (\$6,702,000)
W CAMPUS to FORT HUMBUG 345.0 kV Line	Reconductor line	\$17,959,000	21% (\$3,692,000)
HILLTOP 500-345 kV XFMR	Replace transformer with higher-rated unit	\$31,120,000	21% (\$6,400,000)
LITTLE CHUTE to TALENENERGY MARTINS CRK LLC HARRISBURG 345.0 kV	Reconductor line	\$8,910,000	21% (\$1,831,000)
MIRA LOMA 500-230 kV Transformer Bank	Replace transformer with higher-rated unit	\$33,705,000	100% (\$33,705,000)
BEAVER SUBSTATION to ONANCOCK DIESEL GENERATION FACILITY 230.0 kV L	Reconductor line	\$6,172,000	20% (\$1,268,000)
	Total allocated to project		\$118,670,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.